

From Global to Local: Training-Free Handwritten Digit Classification by Normalized Cross-Correlation

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Key Points:

- Five methods (M1–M5) move from **global** whole-image matching—fixed 2D alignment, 1D raster sliding, or 2D translation—to **local** 7×7 blocks and multi-scale 14×14 regions, all scored by normalized cross-correlation in 1D or 2D space.
- We store 10, 20, or 100 reference images per class without training; classification is template matching only.
- With 100 references, Method 5 raises accuracy from 89.5% to 94.8% (+5.4 pp); with 10 references, from 74.9% to 86.3% (+11.4 pp).

Abstract

We build an effective handwritten-digit classifier without learning any weights. Each image is zero-meaned and ℓ_2 -normalized so that stroke shape, not brightness, drives the match. Five methods (M1–M5) share normalized cross-correlation but search different spaces: fixed 2D alignment, 1D raster sliding, 2D translation, local blocks, and multi-scale blocks. Reference libraries of 10, 20, and 100 templates per class are tested on the same 30,000 MNIST images. With 100 stored examples per digit, accuracy improves from 89.5% (M1) to 94.8% (M5); with only 10 examples per digit, from 74.9% to 86.3%. Unlike deep learning, our classifier never trains weights—it only keeps a small library of reference images (as few as 10 per digit).

Research Background

Handwritten digit recognition on MNIST (LeCun et al., 2010) is widely used to compare classifiers, from template matchers to deep networks (LeCun et al., 1998). Deep models achieve the best accuracy but need large labeled sets and offer limited interpretability. Normalized cross-correlation (NCC) is a standard signal-analysis tool for measuring pattern similarity under shift (Jenkins & Watts, 1968; Lewis, 1995; Brunelli & Poggio, 1993). Applied as template matching, it classifies digits from stored references without training weights (Duda et al., 2001; Jain et al., 2000). This report compares five NCC pipelines (M1–M5) that differ in 1D/2D representation, translation search, and local multi-scale matching, using small reference libraries on MNIST.

1 Introduction

This report builds an effective MNIST classifier from normalized cross-correlation (Lewis, 1995) within a template-matching framework (Brunelli & Poggio, 1993). Each test digit is matched against a stored reference library; the label with the largest correlation score is chosen. Five methods (M1–M5) span **global** whole-image descriptors—fixed 2D alignment, 1D raster sliding, and 2D translation—to **local** block-based descrip-

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tors built from small to large patches at multiple scales. They further differ in **dimensional representation** (1D vector vs. 2D image) and **search region** (fixed alignment, sliding, or local blocks). We evaluate three library sizes—10, 20, and 100 references per class—on the same 30,000 test images. The goal is to create an effective handwritten-digit classifier and to show how cross-correlation in different spaces extracts progressively richer **global** and **local** features without any training step (Duda et al., 2001; Jain et al., 2000).

2 Data and Preprocessing

MNIST provides 28×28 grayscale digit images (LeCun et al., 2010). For each class 0–9 we randomly sample from the official training set 100 reference images and 3,000 test images without overlap (1,000 references and 30,000 tests in total). The same 30,000 test images are reused when the library is reduced to 20 or 10 references per class.

Figure 1 shows one raw sample at pixel resolution (a) and five normalized references per digit (b). Each row contains 28 pixels, and the red grid in (a) marks every cell. After preprocessing, only contrast pattern remains.

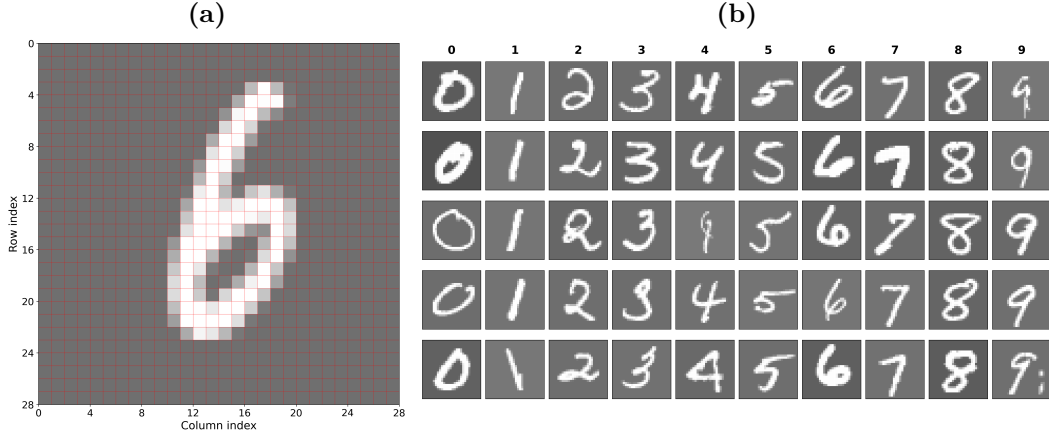


Figure 1. (a) One MNIST digit (28×28): red lines mark pixel boundaries; row and column indices run from 0 to 28. (b) Reference library: five examples per digit 0–9 after zero-mean and ℓ_2 normalization (100 references stored per class in the main experiment).

Each image is a 2D array I . Preprocessing is applied in two steps:

$$I \leftarrow I - \text{mean}(I), \quad I \leftarrow \frac{I}{\|I\|_2}. \quad (1)$$

After this step, only the **shape** of light and dark matters, not absolute brightness.

3 Methods

3.1 Reference r , sample s , and match score ρ

Let r be one reference image and s one test image (both 28×28 after preprocessing). We write R for the cross-correlation function (Jenkins & Watts, 1968); its normalized form follows (Lewis, 1995). The zero-lag autocorrelations are

$$R_{rr}(0) = \sum_{m,n} r(m,n)^2, \quad R_{ss}(0) = \sum_{m,n} s(m,n)^2. \quad (2)$$

If the test image is shifted by (τ_y, τ_x) pixels, the cross-correlation at lag $\boldsymbol{\tau} = (\tau_y, \tau_x)$ is

$$R_{rs}(\boldsymbol{\tau}) = \sum_{m,n} r(m,n) s(m + \tau_y, n + \tau_x). \quad (3)$$

The dimensionless similarity score is

$$\rho(\boldsymbol{\tau}) = \frac{R_{rs}(\boldsymbol{\tau})}{\sqrt{R_{rr}(0) R_{ss}(0)}}, \quad (4)$$

i.e., normalized cross-correlation for template matching (Lewis, 1995; Brunelli & Poggio, 1993). Large ρ means the two patterns are highly similar at that shift. Note that $R_{ss}(0)$ is always computed on the **full** test image and does not change when we slide s . Each test image is compared with every reference; we assign the label of the reference with the largest score.

3.2 Five ways to calculate label score and class

Table 1 and Figure 2 summarize M1–M5; all use Eq. (4) and differ only in representation and search region (details and accuracies in the table).

Table 1. Methods M1–M5: score rules and overall test accuracy (%) at three reference-library sizes (same 30,000 test set). Bottom row: gain of Method 5 over Method 1.

Method	Space	Score computation	Class decision	100 refs	20 refs	10 refs
1	2D tensor	$\rho(\mathbf{0})$ on full image	$\arg \max_r \rho$	89.46	81.56	74.89
2	1D vector	$\max_{\boldsymbol{\tau}} \rho(\boldsymbol{\tau})$ on flattened raster	$\arg \max_r \max \rho$	90.18	82.43	75.35
3	2D tensor	$\max_{\boldsymbol{\tau}} \rho(\boldsymbol{\tau}), \boldsymbol{\tau} \leq 8$	$\arg \max_r \max \rho$	91.81	85.00	76.81
4	2D blocks	$\sum_b \max_{\boldsymbol{\tau}_b} \rho_b$ on 7×7 windows	$\arg \max_r S$	92.51	86.49	82.10
5	Multi-scale 2D	Method 4 at fine + coarse scales; class-wise max	$\arg \max_c \max_r S$	94.81	90.41	86.32
Gain M5–M1 (pp)				5.35	8.84	11.43

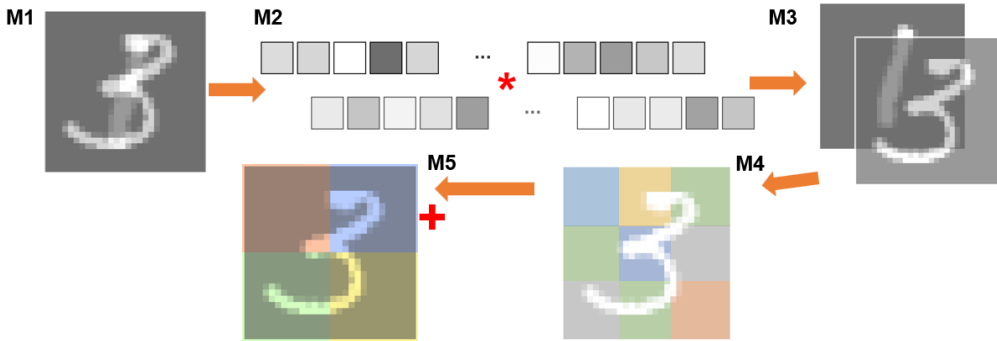


Figure 2. Schematic of Methods 1–5. M2: cross-correlation on a flattened 2D tensor. M3: the reference slides over the sample. M4: the image is divided into small blocks. M5: small and large blocks at multiple scales.

4 Results

All experiments use the same 30,000 test images (3,000 per class). Table 1 and Figure 4 report accuracy for Methods 1–5 at 10, 20, and 100 references per class. Accuracy rises with more references; Methods 2–5 beat Method 1, and this margin widens as the library shrinks (bottom row of Table 1). Figure 3 shows confusion matrices for Methods 1, 3, 4, and 5 at the three library sizes; the diagonal sharpens as references increase and as the method index rises from M1 to M5.

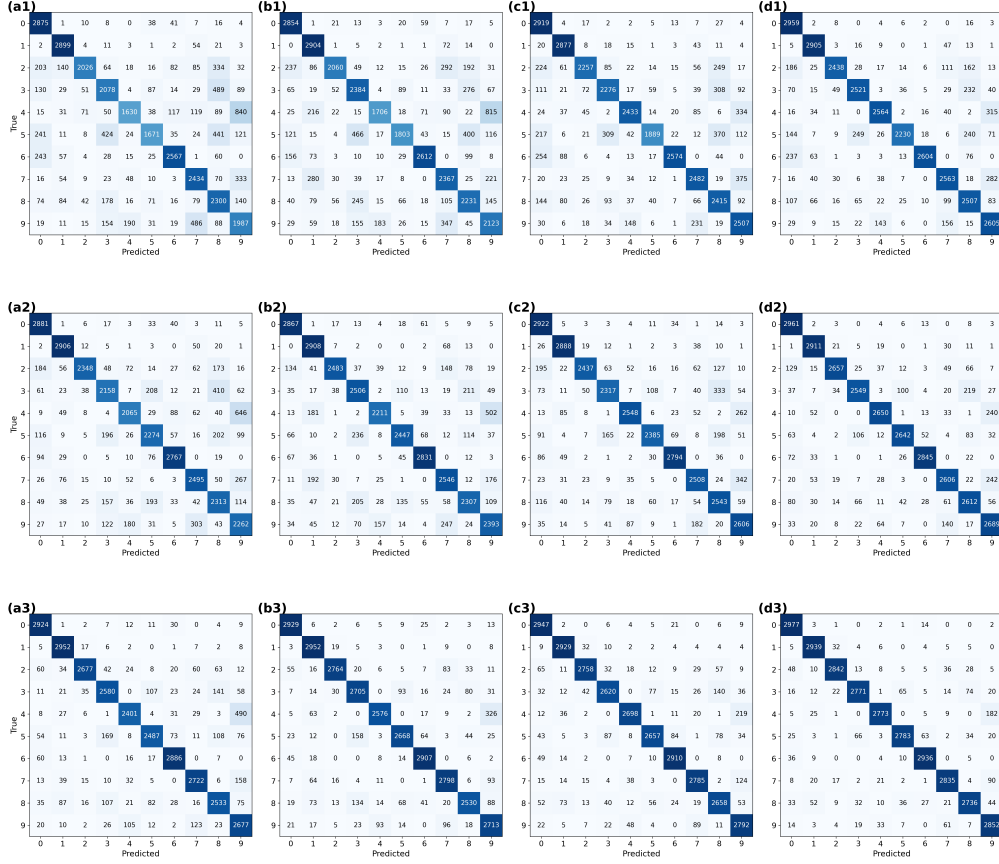


Figure 3. Confusion matrices on the same test set. Row 1 (10 refs): (a1)–(d1) Methods 1, 3, 4, 5; row 2 (20 refs): (a2)–(d2); row 3 (100 refs): (a3)–(d3). Rows within each panel: true label; columns: predicted label.

5 Discussion and Conclusions

Without training any weights, Method 5 reaches 94.8% on 30,000 MNIST test images with 100 references per class and 86.3% with only 10—gains of 5.4 and 11.4 percentage points over Method 1 (Table 1; Figure 4). It also exceeds the ~93% reported for raw-pixel multilayer perceptrons on the same task, while convolutional networks still lead above 99% at the cost of large labeled training sets and learned parameters (LeCun et al., 1998, 2010). Our pipeline stores only a small reference library and scores every decision with explicit normalized cross-correlation.

Methods 1–3 treat the full 28×28 image as one **global** feature. Method 1 uses fixed alignment; Methods 2 and 3 allow 1D and 2D shift to handle positional offset between test and reference digits, but still match the whole pattern at once.

Method 4 brings the largest step in Figure 4(a)—up to 5.29 points over Method 3 at 10 references (Table 1)—even though Methods 2 and 3 already allow 1D and 2D shift. Positional offset therefore cannot be the main bottleneck; recognizing handwritten digits relies chiefly on **local** stroke features, which Method 4 captures by matching each 7×7 block separately. Method 5 combines these **local** cues with **global** layout from 14×14 blocks and a short whole-image shift, so fine detail and digit-level geometry both enter the score. This mix of scales parallels stacked CNN layers—small receptive fields for edges, larger ones for shape (LeCun et al., 1998)—and is why Method 5 ranks highest in Figure 4(a), without learning any weights.

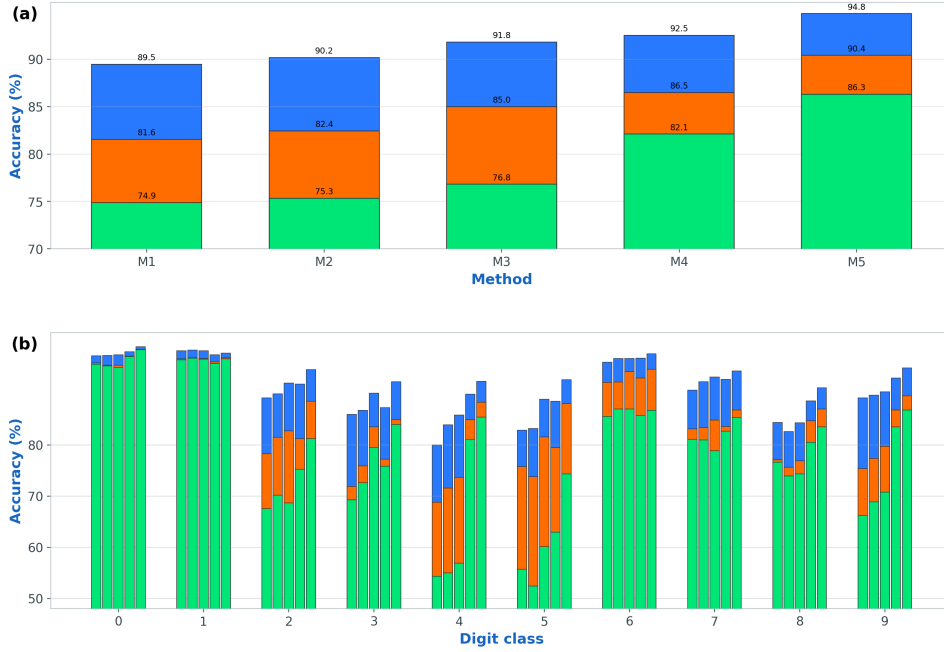


Figure 4. Accuracy on the same 30,000 test images. Layered bars use the same color code in (a) and (b): green = 10 references per class, orange = 20 references per class, blue = 100 references per class. (a) Overall test accuracy vs. method. (b) Per-digit accuracy; within each digit cluster, five method positions (M1–M5, left to right) follow the same scheme.

This staged feature set also explains why smaller libraries benefit more: with few references, a single misaligned global match fails entirely, whereas many local scores can still agree (Figure 4(b), digits 2–5 and 8–9). Moving from one global comparison to localized, multi-scale matching is what makes normalized cross-correlation competitive here without any training.

Acknowledgments

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Appendix A Method 5 prediction examples

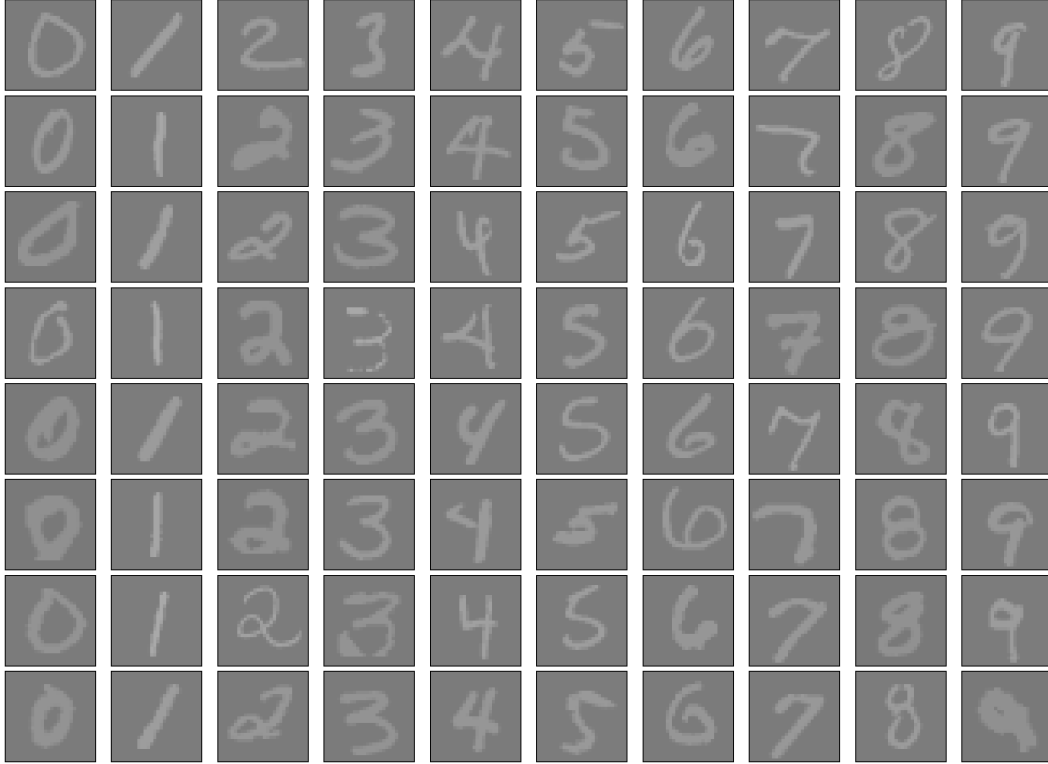


Figure A1. Method 5 correctly classified test samples (100 references per class). Eight rows per digit class; columns are true labels 0–9.

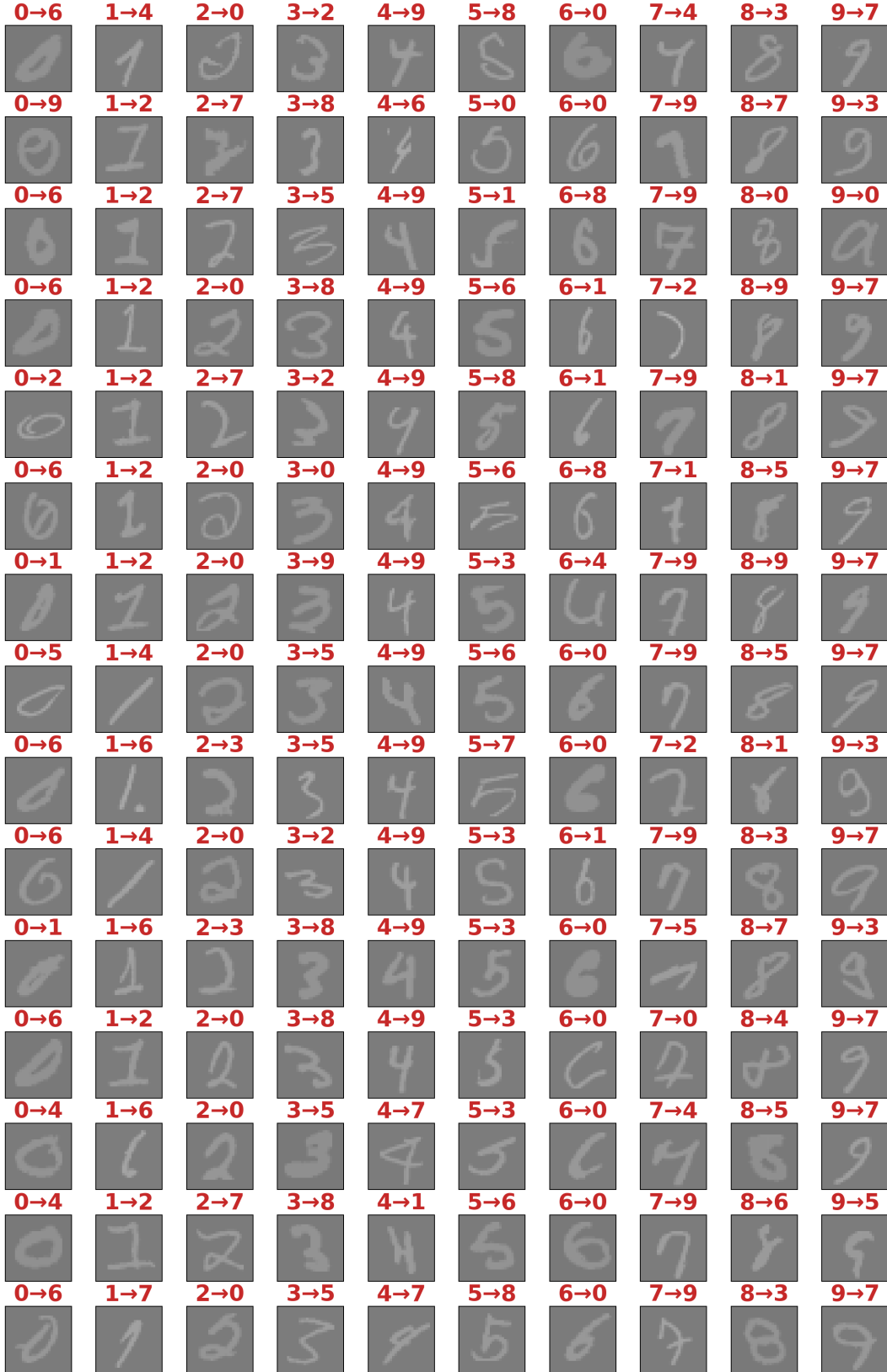


Figure A2. Method 5 misclassified test samples (100 references per class). Each panel shows true → predicted label. Fifteen rows per digit class; columns are true digits 0–9.